28 Chapter Review

Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

SUMMARY OF KEY CONCEPTS

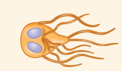
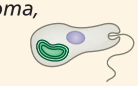
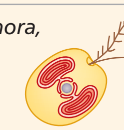
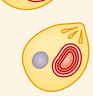
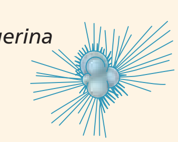
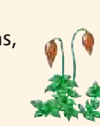
To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkjz9t.

CONCEPT **28.1**

Most eukaryotes are single-celled organisms *(pp. 594–597)*

- Domain Eukarya includes many groups of **protists**, along with plants, animals, and fungi. Unlike prokaryotes, protists and other eukaryotes have a nucleus and other membrane-enclosed organelles, as well as a cytoskeleton that enables them to have asymmetric forms and to change shape as they feed, move, or grow.
- Protists are structurally and functionally diverse and have a wide variety of life cycles. Most are unicellular. Protists include photoautotrophs, heterotrophs, and **mixotrophs**.
- Current evidence indicates that eukaryotes originated by **endosymbiosis** when an archaeal host (or a host closely related to the archaea) engulfed an alpha proteobacterium that would evolve into an organelle found in all eukaryotes, the mitochondrion.
- Plastids are thought to be descendants of cyanobacteria that were engulfed by early eukaryotic cells. The plastid-bearing lineage eventually evolved into **red algae** and **green algae**. Other protist groups evolved from **secondary endosymbiotic** events in which red algae or green algae were themselves engulfed.
- In one hypothesis, eukaryotes are grouped into four supergroups: **Excavata, SAR, Archaeplastida**, and **Unikonta**.

? *Describe similarities and differences between protists and other eukaryotes.*

Key Concept/Eukaryote Supergroup	Major Groups	Key Morphological Characteristics	Specific Examples
CONCEPT **28.2** **Excavates include protists with modified mitochondria and protists with unique flagella** *(pp. 597–601)* ? *What evidence indicates that the excavates form a clade?*	**Diplomonads and parabasalids**	Modified mitochondria	*Giardia, Trichomonas* 
	Euglenozoans Kinetoplastids Euglenids	Spiral or crystalline rod inside flagella	*Trypanosoma, Euglena* 
CONCEPT **28.3** **SAR is a highly diverse group of protists defined by DNA similarities** *(pp. 601–609)* ? *Although they are not photosynthetic, apicomplexan parasites such as* Plasmodium *have modified plastids. Describe a current hypothesis that explains this observation.*	**Stramenopiles** Diatoms Oomycetes Brown algae	Hairy and smooth flagella	*Phytophthora, Laminaria* 
	Alveolates Dinoflagellates Apicomplexans Ciliates	Membrane-enclosed sacs (alveoli) beneath plasma membrane	*Pfiesteria, Plasmodium, Paramecium* 
	Rhizarians Radiolarians Forams Cercozoans	Amoebas with threadlike pseudopodia	*Globigerina* 
CONCEPT **28.4** **Red algae and green algae are the closest relatives of plants** *(pp. 609–611)* ? *On what basis do systematists place plants in the same supergroup (Archaeplastida) as red and green algae?*	**Red algae**	Phycoerythrin (photosynthetic pigment)	*Porphyridium, Palmaria* 
	Green algae	Plant-type chloroplasts	*Chlamydomonas, Ulva* 
	Plants	(See Chapters 29 and 30.)	Mosses, ferns, conifers, flowering plants 
CONCEPT **28.5** **Unikonts include protists that are closely related to fungi and animals** *(pp. 611–614)* ? *Describe a key feature for each of the main protist subgroups of Unikonta.*	**Amoebozoans** Tubulinids Slime molds Entamoebas	Amoebas with lobe-shaped or tube-shaped pseudopodia	*Amoeba, Dictyostelium* 
	Opisthokonts	(Highly variable; see Chapters 31–34.)	Choanoflagellates, nucleariids, animals, fungi